

It has been growing ever since. A survey made in the spring of 1956, published in *Business Week* and a number of other McGraw-Hill trade publications, indicated that in 1953 private corporate expenditures for research and development were about \$3.7 billion. By 1956 they had grown to \$5.5 billion and present corporate planning called for this to be running at the rate of better than \$6.3 billion by 1959. Equally startling, this survey indicated that by 1959, or in just three years, a number of our leading industries expect to get from 15 per cent to more than 20 per cent of their total sales from products which were not in commercial existence in 1956.

In the spring of 1957 the same source made a similar survey. If the totals revealed in 1956 were startling in their significance, those revealed just one year later might be termed explosive. Research expenditures were up 20 per cent from the previous year's total to \$7.3 billion! This represents almost a 100 per cent growth in four years. It means the actual growth in twelve months was \$1 billion more than only a year before had been expected as the total growth that would occur in the ensuing thirty-six months.

Meanwhile, anticipated research expenditures in 1960 were estimated at \$9 billion! Furthermore, all manufacturing industries, rather than just a few selected industries as represented in the earlier survey, expected that 10 per cent of 1960 sales would be from products not yet in commercial existence only three years before. For certain selected industries, this percentage—from which sales representing merely new model and style changes had been excluded—was several times higher.

The impact of this sort of thing on investment can hardly be over-stated. The cost of this type of research is becoming so great that the corporation which fails to handle it wisely from a commercial standpoint may stagger under a crushing burden of operating expense. Furthermore, there is no quick and easy yardstick for either management or the investor to measure the profitability of research. Just as even the ablest professional baseball player cannot expect to get a hit much more often than one out of every three times he comes to bat, so a sizable number of research projects, governed merely by the law of averages, are bound to produce nothing profitable at all. Furthermore, by pure chance, an abnormal number of such unprofitable projects may happen to be bunched together in one particular span of time in even the best-run commercial laboratory. Finally, it is apt to take from seven to eleven years from the time a project is first conceived